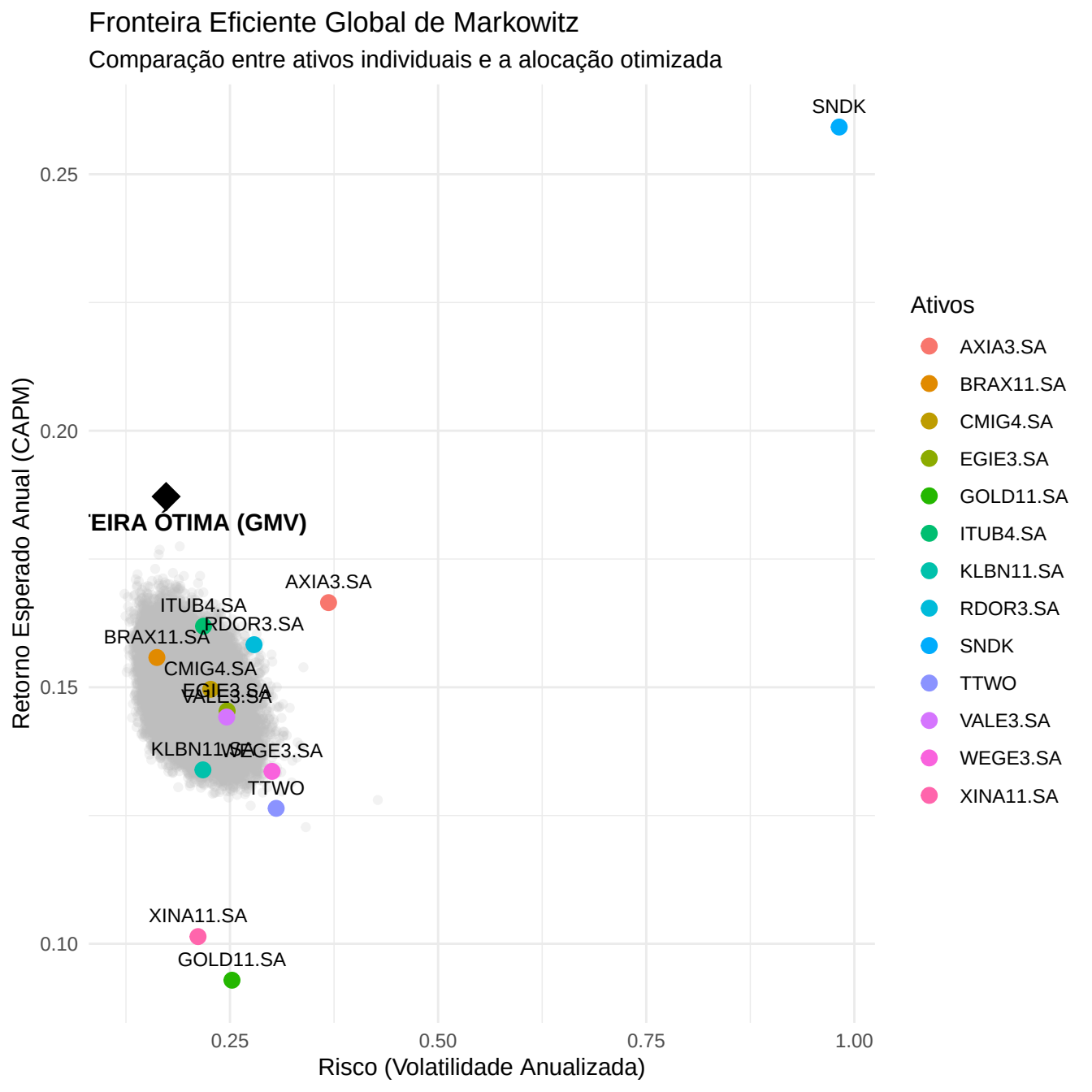
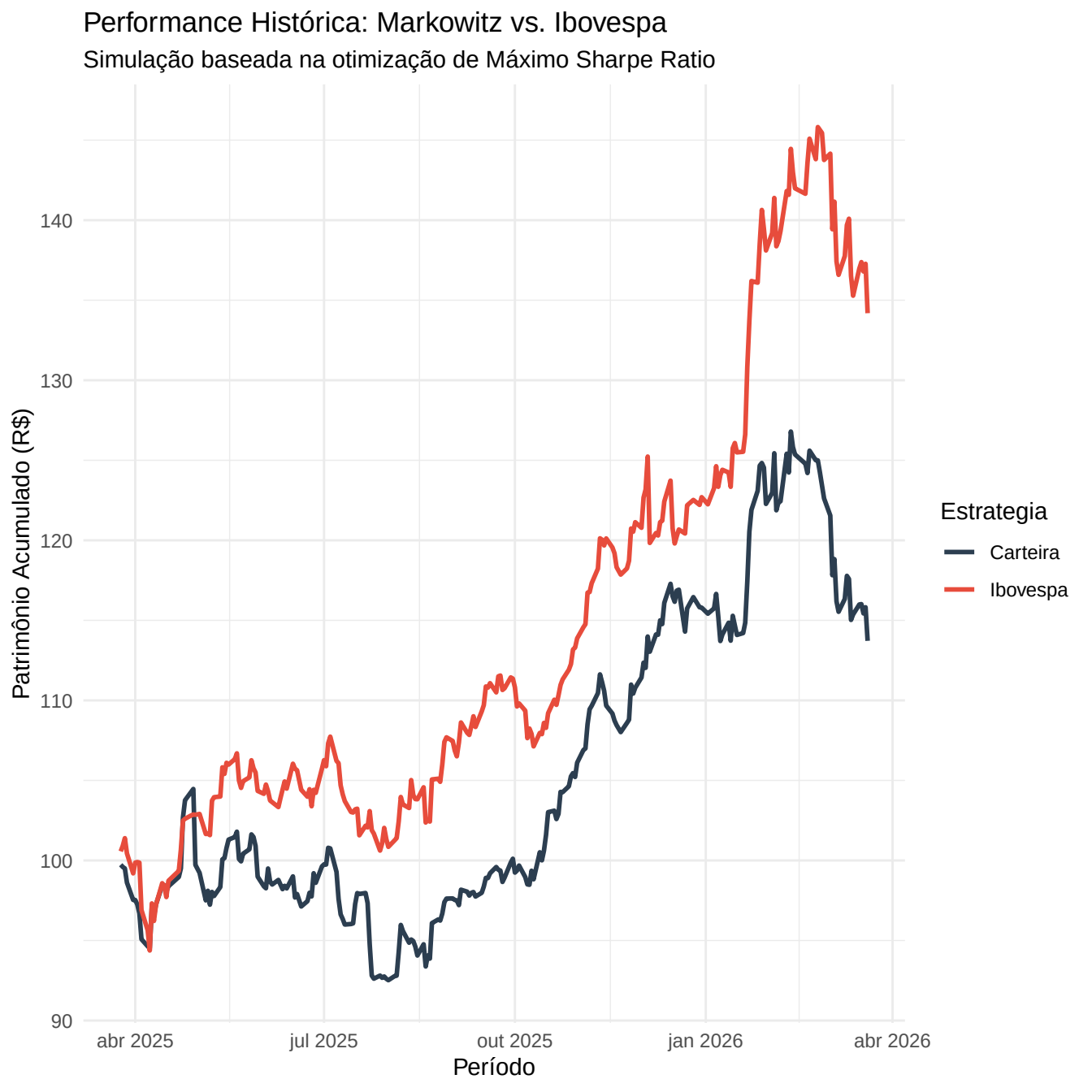

Fronteira Eficiente Global de Markowitz
Comparação entre ativos individuais e a alocação otimizada
SNDK
0.25
0.20
0.15
0.10
EIRA OTIMA (GMV)
AXIA3.SA
ITUB4.SA
RDOR3.SA
BRAX11.SA
CMIG4.SA
EGIE3.SA
VALE3.SA
KLBN11.SA
WEGE3.SA
TTWO
XINA11.SA
GOLD11.SA
Retorno Esperado Anual (CAPM)
0.25
0.50
0.75
1.00
Risco (Volatilidade Anualizada)
Ativos
AXIA3.SA
BRAX11.SA
CMIG4.SA
EGIE3.SA
GOLD11.SA
ITUB4.SA
KLBN11.SA
RDOR3.SA
SNDK
TTWO
VALE3.SA
WEGE3.SA
XINA11.SA


Performance Histórica: Markowitz vs. Ibovespa
Simulação baseada na otimização de Máximo Sharpe Ratio
Patrimônio Acumulado (R$)
140
130
120
110
100
90
abr 2025
jul 2025
out 2025
jan 2026
abr 2026
Período
Estrategia
Carteira
Ibovespa

Matriz de Dependência Linear entre Ativos

Calculada sobre os log−retornos diários

	AXIA3.SA	BRAX11.SA	CMIG4.SA	EGIE3.SA	GOLD11.SA	ITUB4.SA	KLBN11.SA	RDOR3.SA	SNDK	TTWO	VALE3.SA	WEGE3.SA	XINA11.SA
XINA11.SA	0.12	0.07	0.11	−0.01	0.22	0.17	−0.02	0.2	0.18	−0.15	0.08	0.25	1
WEGE3.SA	0	0.27	0.27	0.35	0.02	0.28	0.12	0.04	0.45	−0.5	0.29	1	0.25
VALE3.SA	0.43	0.62	0.57	0.35	0.26	0.47	0.64	0.27	0.45	−0.11	1	0.29	0.08
TTWO	0.27	0.17	0.03	0.08	−0.12	0.25	0.36	0.02	−0.08	1	−0.11	−0.5	−0.15
SNDK	0.62	0.7	0.47	0.58	0.04	0.62	0.52	0.33	1	−0.08	0.45	0.45	0.18
RDOR3.SA	0.58	0.64	0.69	0.44	−0.23	0.71	0.33	1	0.33	0.02	0.27	0.04	0.2
KLBN11.SA	0.75	0.81	0.62	0.58	0.22	0.62	1	0.33	0.52	0.36	0.64	0.12	−0.02
ITUB4.SA	0.76	0.87	0.81	0.72	−0.18	1	0.62	0.71	0.62	0.25	0.47	0.28	0.17
GOLD11.SA	0.17	0.1	−0.23	0.09	1	−0.18	0.22	−0.23	0.04	−0.12	0.26	0.02	0.22
EGIE3.SA	0.75	0.81	0.73	1	0.09	0.72	0.58	0.44	0.58	0.08	0.35	0.35	−0.01
CMIG4.SA	0.71	0.77	1	0.73	−0.23	0.81	0.62	0.69	0.47	0.03	0.57	0.27	0.11
BRAX11.SA	0.89	1	0.77	0.81	0.1	0.87	0.81	0.64	0.7	0.17	0.62	0.27	0.07
AXIA3.SA	1	0.89	0.71	0.75	0.17	0.76	0.75	0.58	0.62	0.27	0.43	0	0.12

Correlação (Pearson)

1.0
0.5
0.0
−0.5
−1.0